

COVER
COVER.SchDoc

OVERVIEW
OVERVIEW.SchDoc

EXPANSION
EXPANSION.SchDoc

PORTENTA X8
PORTENTAX8.SchDoc

CAN FD
CANFD.SchDoc

RS485
RS485.SchDoc

IMU
IMU.SchDoc

EEPROM
EEPROM.SchDoc

1000BASE-T
1000BASE-T.SchDoc

SUPPLY
SUPPLY.SchDoc

10BASE-T1S
10BASE-T1S.SchDoc

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go-pika-spark-robot-bg-transparent.png.
File does not exist.

PCB100
Printed Circuit Board (Bare)
PCB-BARE

Pass100



Fiducial

Pass101



Fiducial

Pass102



Fiducial

Pass103



Fiducial

Pass104



Fiducial

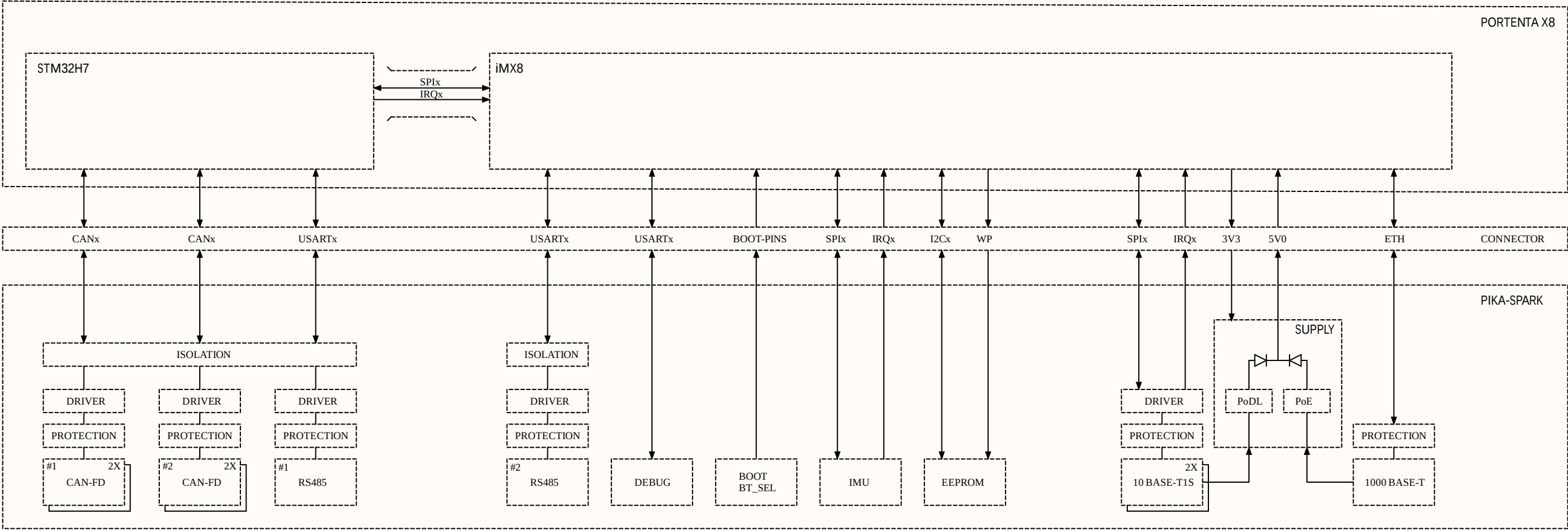
Pass105

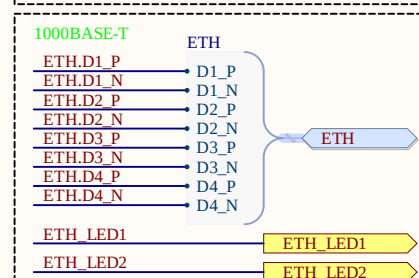


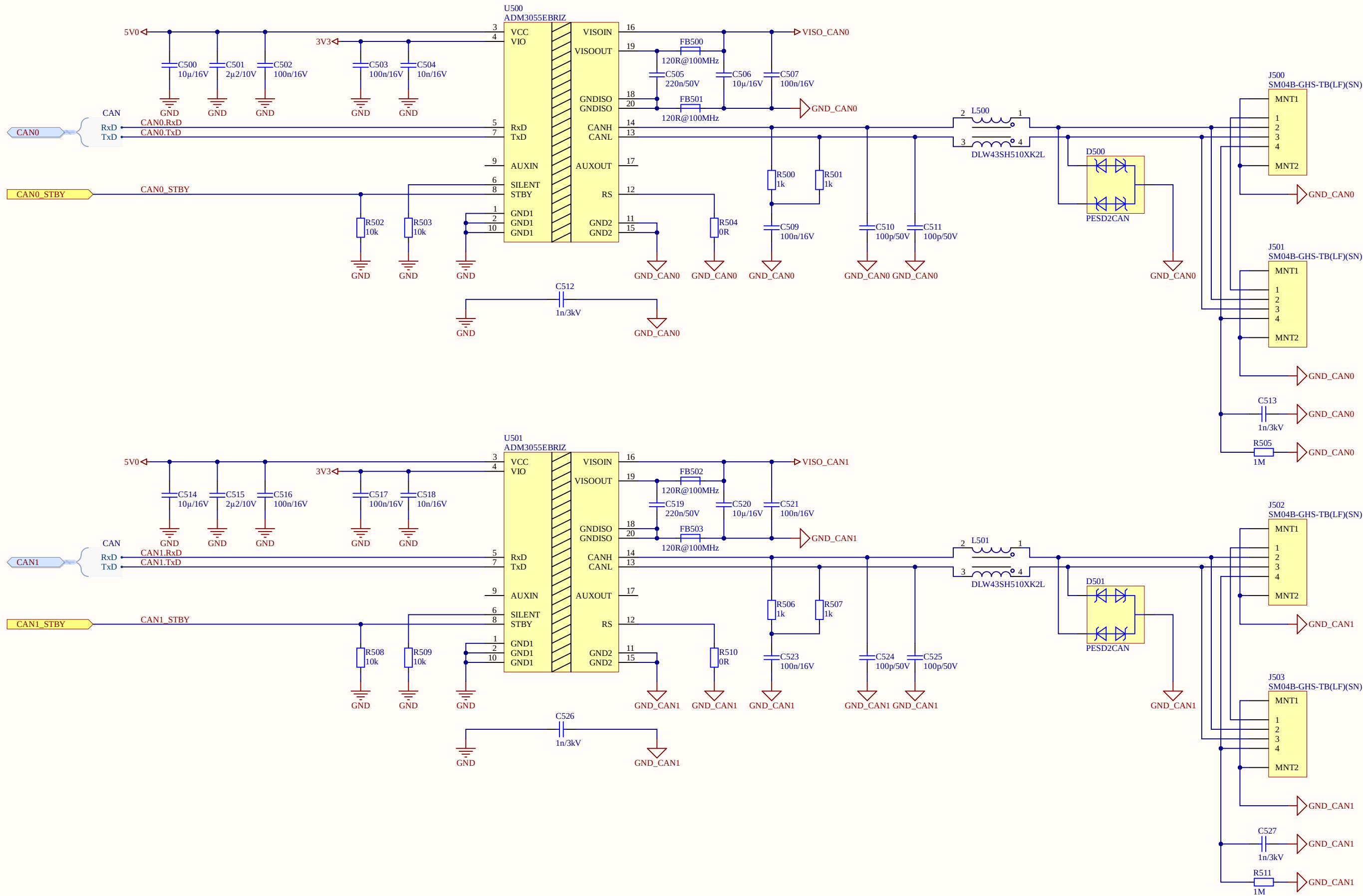
Fiducial

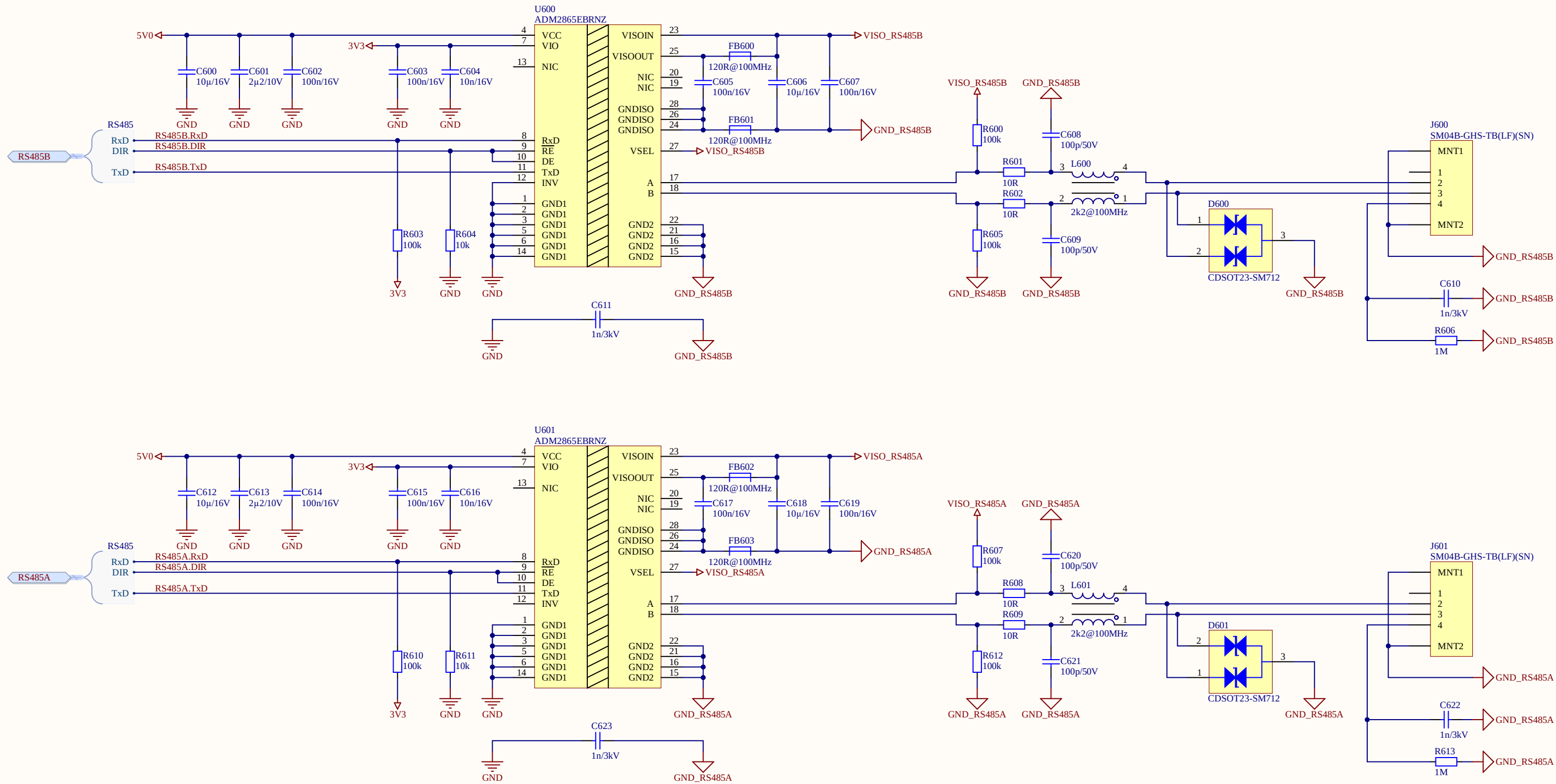
REV	DESCRIPTION	DATE	BY
V1	...	01/02/23	LRUHMI

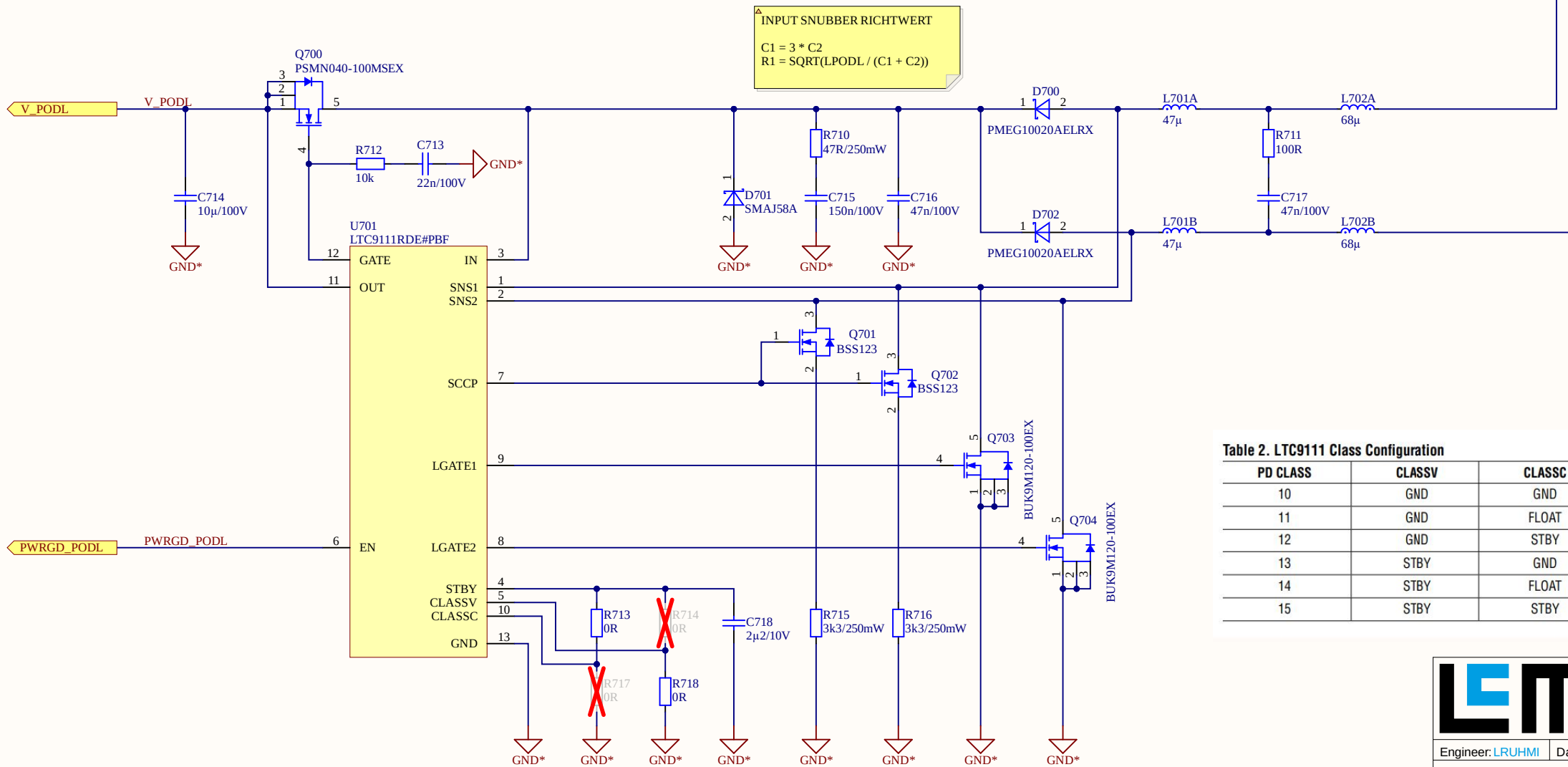
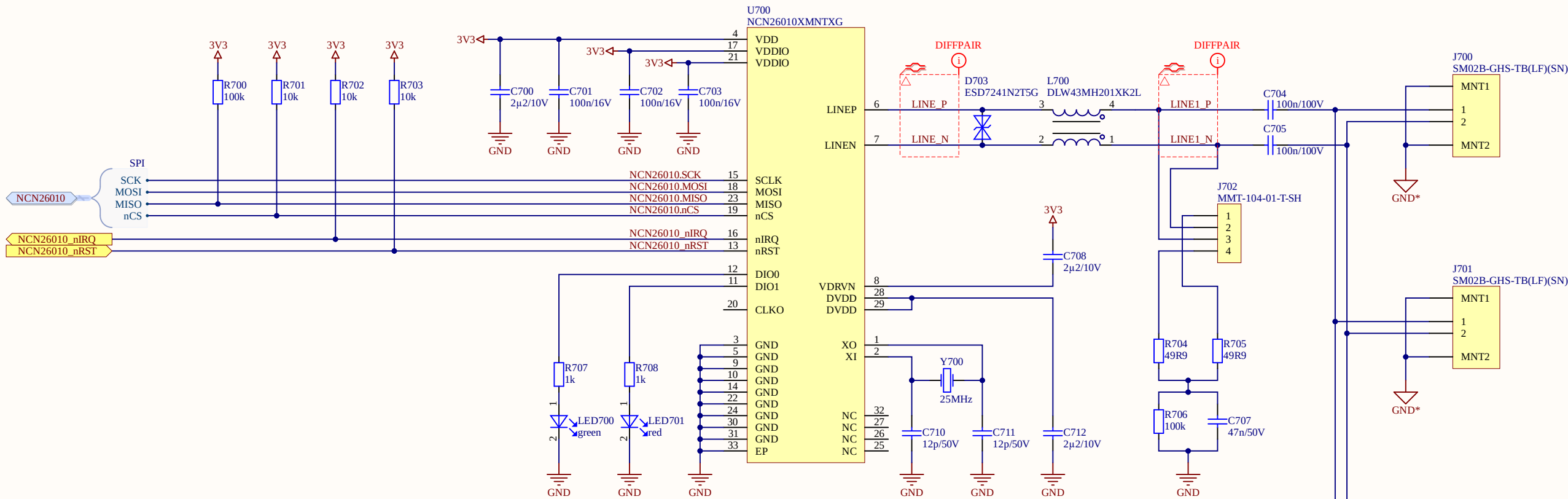
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INPUT SNUBBER RICHTWERT
 $C1 = 3 * C2$
 $R1 = \sqrt{L_{PODL} / (C1 + C2)}$

Table 2. LTC9111 Class Configuration

PD CLASS	CLASSV	CLASSC
10	GND	GND
11	GND	FLOAT
12	GND	STBY
13	STBY	GND
14	STBY	FLOAT
15	STBY	STBY

Table 1. PSE and PD Valid Class Pairs

PSE CLASS	MAX PD POWER (W)	PD VOLTAGE (MIN) (V)	PD VOLTAGE (MAX) (V)	I _{MAX} (mA)	COMPATIBLE PD CLASS(ES)
10	1.23	14	30	92	10
11	3.2	14	30	240	10, 11
12	8.4	14	30	632	10, 11, 12
13	7.7	35	58	231	13
14	20	35	58	600	13, 14
15	52	35	58	1579	13, 14, 15



LINZ
CENTER OF
MECHATRONICS
GMBH

PIKA-SPARK
10BASE-T1S

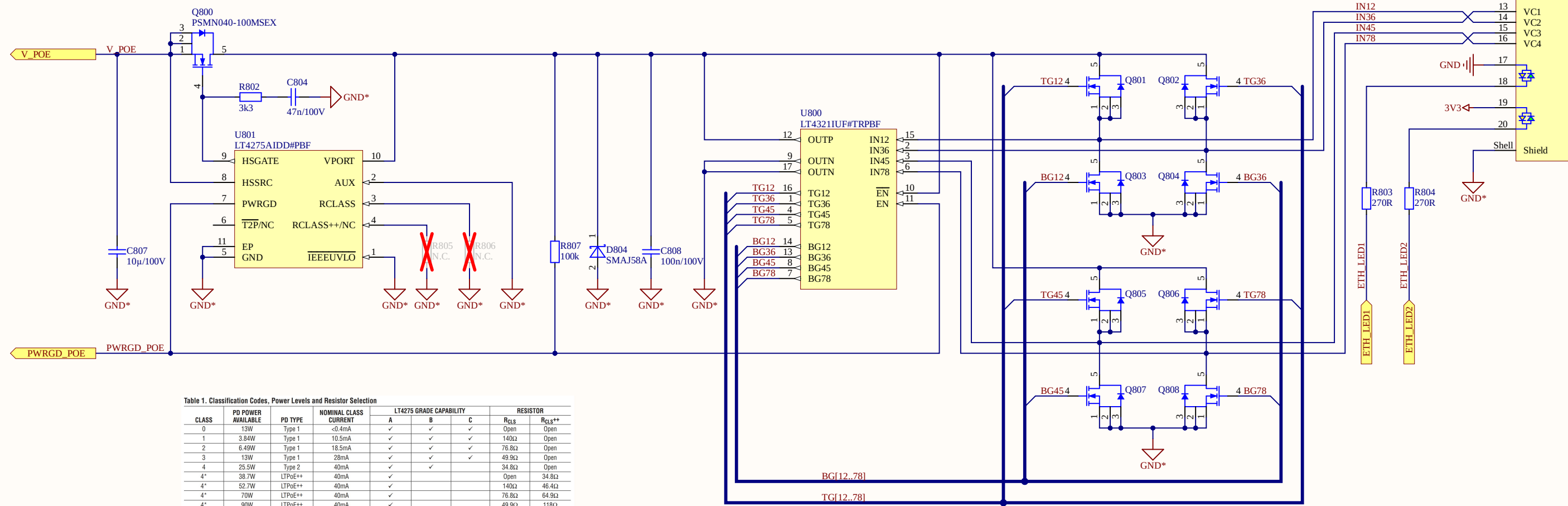
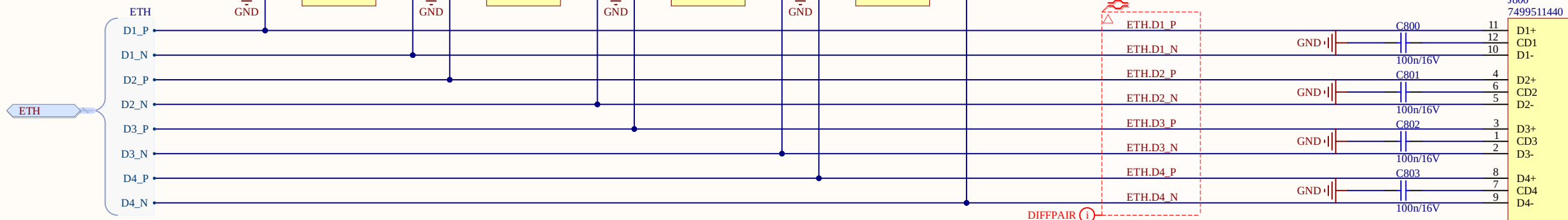
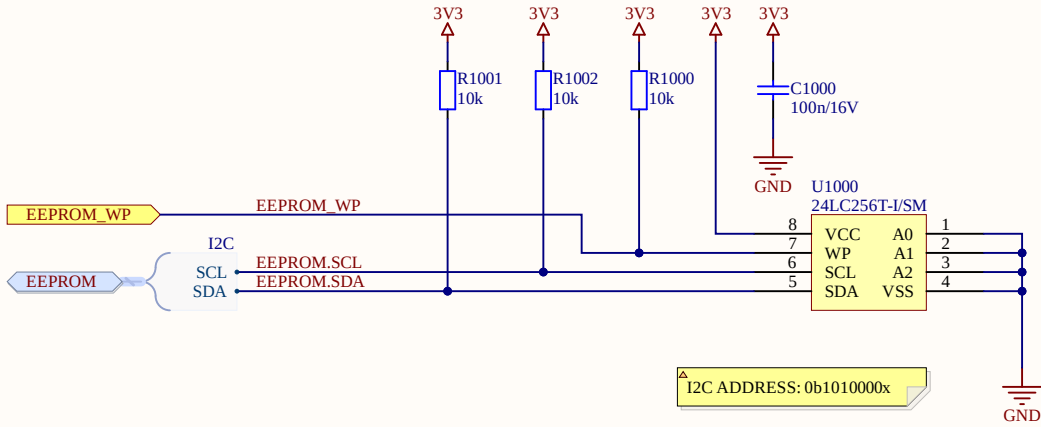


Table 1. Classification Codes, Power Levels and Resistor Selection

CLASS	PD POWER AVAILABLE	PD TYPE	NOMINAL CLASS CURRENT	LT4275 GRADE CAPABILITY			RESISTOR	
				A	B	C	R _{GLS}	R _{GLS} ++
0	13W	Type 1	<0.4mA	✓	✓	✓	Open	Open
1	3.84W	Type 1	10.5mA	✓	✓	✓	140Ω	Open
2	6.49W	Type 1	18.5mA	✓	✓	✓	76.8Ω	Open
3	13W	Type 1	28mA	✓	✓	✓	49.9Ω	Open
4	25.5W	Type 2	40mA	✓	✓		34.8Ω	Open
4*	38.7W	LTPOE++	40mA	✓			Open	34.8Ω
4*	52.7W	LTPOE++	40mA	✓			140Ω	46.4Ω
4*	70W	LTPOE++	40mA	✓			76.8Ω	64.9Ω
4*	90W	LTPOE++	40mA	✓			49.9Ω	118Ω

*An LTPOE++ PD will be classified as class 4 by an IEEE 802.3 compliant PSE.



$V_{\text{Zener}} \leq 145\text{V} - V_{\text{INMAX}}$
 $V_{\text{Zener}} \leq 145\text{V} - 60\text{V} = 85\text{V}$ [1.5W] gewählt 68V

$R_{\text{FB}} = R_{\text{REF}} * \text{NPS} * (V_{\text{OUT}} + V_{\text{F}}) / V_{\text{REF}}$
 $R_{\text{FB}} = 10\text{k} * 6 * (5\text{V} + 0.3\text{V}) / 1.0\text{V} = 318\text{k}$

$R_{\text{FBnew}} = V_{\text{OUT}} / V_{\text{OUTmeasured}} * R_{\text{FB}}$
 $R_{\text{FBnew}} = 309\text{k}$

